Now, comparison operations can drive logic, but they are the simplest kind of logical expression: a single predicate – a single property or relationship. `$a > $b` is a predicate that asserts that `$a` is greater than `$b` (and can be true or false). More complex logical expressions consist of multiple predicates and require that the predicates themselves are connected by special operators.

Logical Operators

If we have two predicates in an expression, then we need to reduce them to a single value to drive a control structure such as `if` or `while`. Let's call our predicates A and B. We could decide that the overall expression is `true` if both A and B are `true`, or either is `true`, or even if only one is `true`. Here are the logical operators that allow predicates to be joined together in these ways:

Table 4-4. *Logical operators*

`&&` `and`	Logical "AND", `true` if both operands are `true`
`||` `or`	Logical "OR", `true` if either operand is `true`
`xor`	Logical "EXCLUSIVE OR", `true` if only one operand is `true`, but not both
`!`	Logical "NOT", `true` if the operand is `false`

Logical `AND` and `OR` have two representations in PHP because of something called operator precedence, which we'll get to in due course. Firstly, let's look at some simple examples:

```
var_dump(true and true); // true
var_dump(true && false); // false
var_dump(false or true); // true
var_dump(false xor true); // true
var_dump(true xor false); // true
var_dump(true xor true); // false
var_dump(!false); // true
```